n-dimensional Boolean algebras

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Outline

1. Church algebras and the λ-calculus
2. n-dimensional Boolean algebras
3. n-dimensional propositional calculus
4. Rewriting n-PC formulas

Semantic completeness

The problem

Given a class of models of the λ-calculus, is it the case that *any* λ-theory is representable in it ?

Some (ad hoc) incompleteness results :

- Sott semantics (Honsell and Ronchi della Rocca 1992),
- Stable semantics (Gouy 1995),
- Strongly stable semantics (Bastonero 1998).

A *general* incompleteness result, based on Universal Algebra (Manzonetto and Salibra 2006) : *The λ-models arising from cpo-based semantics are* indecomposable*, and there exist λ-theories admitting only* decomposable *models*.

Church Algebras

An algebra **A** is a *Church algebra* if it admits a ternary operator $q(x, y, z)$ and two constants $0, 1$ such that the following identities are satisfied :

$$q(1, x, y) = x \qquad q(0, x, y) = y$$

Examples :

- Rings with unit : $q(x, y, z) = xy + (1 - x)z$.
- Heyting algebras : $q(x, y, z) = (x \vee z) \wedge ((x \to 0) \vee y)$.
- Combinatory algebras : $q(x, y, z) = (xy)z; \quad 1 = \mathbf{k}; \quad 0 = \mathbf{sk}$.
- Lambda abstraction algebras : $q(x, y, z) = (xy)z \quad 1 = \lambda xy.x; \quad 0 = \lambda xy.y$.

Factor congruences and decomposition

Decomposition

$$\mathbf{A} \cong \mathbf{B} \times \mathbf{C}$$
$$\Updownarrow$$
$$\exists \theta, \overline{\theta} \text{ such that } \mathbf{B} \cong \mathbf{A}/\theta \text{ and } \mathbf{C} \cong \mathbf{A}/\overline{\theta}$$
$$\Updownarrow$$
$$\exists \theta, \overline{\theta} \text{ as above, and moreover } \theta \cap \overline{\theta} = \Delta \text{ and } \theta \circ \overline{\theta} = \nabla$$

$(\theta, \overline{\theta})$ as in the third line above is called *a pair of complementary factor congruences*.

Fact

If $(\theta, \overline{\theta})$ is a CFC-pair, then the map

$$(x, y) \mapsto \text{ The unique } z \text{ such that } \ x\theta z\overline{\theta}y$$

is a homomorphism from $\mathbf{A} \times \mathbf{A}$ into $\mathbf{A}$

An algebra $\mathbf{A}$ is

- *directly indecomposable* if it does not admit any non trivial CFC-pair.
- *simple* if $Con(\mathbf{A}) = \{\Delta, \nabla\}$.

Central elements

Definition

An element e of a double-pointed algebra is *central* if $(\theta(1,e),\theta(0,e))$ is a CFC-pair.

Within Church Algebras

$$\begin{array}{rcl} \text{The BA of central elements} & \rightleftarrows & \text{The BA of CFC-pairs} \\ e \in \mathrm{Ce}(\mathbf{A}) & \mapsto & (\theta(1,e),\theta(0,e)) \\ (\theta,\overline{\theta}) & \mapsto & \text{The unique } e \text{ such that } 1\theta e\overline{\theta}0 \end{array}$$

BA-operations on central elements

$$x \wedge y = q(x,y,0); \quad x \vee y = q(x,1,y); \quad \neg x = q(x,0,1)$$

Equational Characterization of the central element e

1. $q(e,x,x) = x$
2. $q(e,q(e,x,y),z) = q(e,x,z) = q(e,x,q(e,y,z))$
3. $q(e,f(\overline{x}),f(\overline{y})) = f(q(e,x_1,y_1),\ldots,q(e,x_n,y_n))$ (for every f in the type)
4. $q(e,1,0) = e$

Non-trivial central elements in λ-models

The λ-term $\Omega = (\lambda x.xx)(\lambda x.xx)$ is an *easy term* : it can be consistently equated to any other closed λ-term.

Fact

In the term model of the λ-theory $\theta(1, \Omega) \cap \theta(0, \Omega)$, where $1 = \lambda xy.x, 0 = \lambda xy.y$, Ω is a *non-trivial* central element (hence the term model is decomposable).

Lemma

Indecomposable lambda-algebras are closed by subalgebras.

Corollary

Any model of the λ-theory $\theta(1, \Omega) \cap \theta(0, \Omega)$ is decomposable (since if $\mathcal{M}$ is a model of the λ-calculus then the term model of $\mathrm{Th}(\mathcal{M})$ is a subalgebra of $\mathcal{M}$).

The Scott semantics is incomplete

Scott λ-algebras are simple, hence indecomposable ("Scott is always simple")

Sketch of the proof :

- Let $\phi \neq \Delta$ be a congruence on a Scott λ-algebra $\mathbf{A}$. Suppose $a\phi b$ with $a \not\leq b$.
- Given $c \in \mathbf{A}$, $g_c(x) = \begin{cases} c & \text{if } x \not\leq b \\ \perp & \text{otherwise} \end{cases}$ is continuous.
- Let $d \in \mathbf{A}$ representing g_c (for all x, $dx = g_c(x)$).
- Hence $c = da \; \phi \; db = \perp$, and $\phi = \nabla$

Theorem

The Scott semantics of the λ-calculus is incomplete.

(no Scott model may have $\theta(1, \Omega) \cap \theta(0, \Omega)$ as theory)

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Algebras of dimension n

An algebra **A** is an *algebra of dimension n* (n-A) if it admits a $(n+1)$-ary operator $q(x, y_1, \ldots, y_n)$ and n constants $e_1, \ldots, e_n$ such that for all $1 \leq i \leq n$ the following identity is satisfied :

$$q(e_i, y_1, \ldots y_n) = y_i$$

Examples :

- Lambda abstraction algebras :
  - $e_i = \lambda x_1 \ldots x_n . x_i$
  - $q(x, y_1, \ldots, y_n) = x y_1 \ldots y_n$.
- n-sets on a set U : $\{(X_1, \ldots, X_n) : X_i \subseteq U\}$;
  - $e_i = (\underbrace{\emptyset, \ldots, \emptyset}_{(i-1)\times}, U, \emptyset, \ldots, \emptyset)$,
  - $q(X, Y^1, \ldots, Y^n) = (\bigcup_{j=1}^n X_j \cap Y_1^j, \ldots, \bigcup_{j=1}^n X_j \cap Y_n^j)$.

n-central elements

If $\mathbf{A}$ is an n-DA, then $c \in A$ is called *n-central* if the sequence $(\theta(c, e_1), \ldots, \theta(c, e_n))$ is an n-tuple of complementary factor congruences of $\mathbf{A}$, meaning that :

- $\bigcap_{i \leq n} \theta(c, e_i) = \Delta$;
- for all $a_1, \ldots, a_n \in A$, $q(c, a_1, \ldots, a_n)$ is the unique element such that $a_i \; \theta(c, e_i) \; q(c, a_1, \ldots, a_n)$ for all $1 \leq i \leq n$;

.

If c is n-central, then $\mathbf{A} \cong \mathbf{A}/\theta(c, e_1) \times \ldots \times \mathbf{A}/\theta(c, e_n)$.

Equational characterisation

$c \in \mathbf{A}$ is n-central iff :

B1 $q(c, e_1, \ldots, e_n) = c$.

B2 $q(c, x, x, \ldots, x) = x$ for every $x \in A$.

B3 If $\sigma \in \nu$ has arity k, then

$$\begin{aligned} & q(c, \sigma(x_{11}, \ldots, x_{1k}), \ldots, \sigma(x_{n1}, \ldots, x_{nk})) \\ = \; & \sigma(q(c, x_{11}, \ldots, x_{n1}), \ldots, q(c, x_{1k}, \ldots, x_{nk})). \end{aligned}$$

n-BAs

Proposition

The set of n-central elements of an n-DA **A** is a subalgebra of the pure reduct of **A** (i.e. : if $x, y_1, \ldots, y_n$ are n-central, then $q(x, y_1, \ldots, y_n)$ is n-central.)

A *Boolean algebra of dimension n* (n-BA) is an n-DA all of whose elements are n-central.

Examples :

- The pure n-BA $(\{e_1, \ldots, e_n\}, q, e_1, \ldots, e_n)$ of *generalized truth values*.
- n-partitions of set U : $\{(X_1, \ldots, X_n) : X_i \subseteq U,\ \bigcup_i X_i = U,\ X_i \cap X_j = \emptyset\}$;

BAs and n-BAs

A first representation theorem

Any pure n-BA is isomorphic to a subalgebra of the n-BA of the n-partitions of a set X.

The 2-elements Boolean algebra is *primal* : all finite Boolean functions are definable by AND, OR, NOT. This property is inherited by n-BAs.

Primality in n-BAs

Let $\mathbf{A}$ be a finite τ-algebra of cardinality n. Then $\mathbf{A}$ is primal if and only if it is an n-BA.

A second representation theorem

The inner Boolean algebra of an n-BA

Let $\mathbf{A}$ be an n-BA. The set $B_{\mathbf{A}} = \{x \in A \ : \ q(x, y, y, e_3, \ldots, e_n) = y\}$ with the operations :

$$x \wedge y = q(x, e_1, y, e_3, \ldots, e_n); \quad x \vee y = q(x, y, e_2, e_3, \ldots, e_n);$$

$$\neg x = q(x, e_2, e_1, e_3 \ldots, e_n); \quad 0 = e_1; \quad 1 = e_2,$$

is a Boolean algebra, written $\mathbf{B_A}$.

Intuition : the elements of $B_{\mathbf{A}}$ correspond to the n-partitions of the form $(X, U \setminus X, \emptyset, \ldots, \emptyset)$, for $X \subseteq U$.

Theorem

Any n-BA $\mathbf{A}$ is isomorphic to the n-BA of n-central elements of the Boolean vector space $\mathbf{B_A}^n$.

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n-dimensional propositional calculus

Formulas : $\Phi, \Psi := x \in Var \mid e_i \in \mathbf{n} = \{e_1, \ldots, e_n\} \mid q(\Phi, \Psi_1, \ldots, \Psi_n)$
Interpretation : given $\rho : Var \to \mathbf{n}$

- $[x]_\rho = \rho(x)$
- $[e_i]_\rho = e_i$
- $[q(\Phi, \Psi_1, \ldots, \Psi_n)]_\rho = [\Psi_j]_\rho$ where $[\Phi_\rho] = e_j$

(i-)Validity :
$\Gamma \models_{(\mathbf{n}, e_i)} \Delta$ if for all $\rho : Var \to \mathbf{n}$:

if for all $\Phi \in \Gamma$ $[Phi]_\rho = e_i$ then there exists $\Psi \in \Delta$ such that $[\Psi]_\rho = e_i$

Simmetry of truth values

$\Gamma \models_{(\mathbf{n}, e_i)} \Delta$ iff $\Gamma^{(i,j)} \models_{(\mathbf{n}, e_j)} \Delta^{(i,j)}$

e_n is chosen as designated value.

Encoding of tabular logics into n-PC

Theorem

All tabular logic $\models_{(V,1)}$ may be faithfully encoded into n-PC.

Example : the 3-valued Post logic

- $V = \{0, \frac{1}{2}, 1\}$, $C = \{\wedge, \vee, \neg\}$
- $x \wedge y = min(x, y)$, $x \vee y = max(x, y)$
- $\neg 0 = 1, \neg \frac{1}{2} = 0, = 1, \neg 1 = \frac{1}{2}$

Encoding into 3-PC :

- $0^* = e_1, \frac{1}{2}^* = e_2, 1^* = e_3$.
- $(x \wedge y)^* = q(x^*, e_1, q(y^*, e_1, e_2, e_2), y^*)$
- $(x \vee y)^* = q(x^*, y^*, q(y^*, e_2, e_2, e_3), e_3)$
- $(\neg x)^* = q(x^*, e_3, e_1, e_2)$.

Fact

$$\models_{(V,1)} \Phi \Leftrightarrow \models_{(\{e_1,e_2,e_3\},e_3)} \Phi^*$$

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Rewriting n-PC formulas

(h_0) $q(e_i, x_1, \ldots, x_n) \rightarrowtail_{\text{hnf}} x_i$

(h_1) $q(q(x, y_1 \ldots, y_n), z_1, \ldots, z_n) \rightarrowtail_{\text{hnf}}$
$q(x, q(y_1, z_1, \ldots, z_n), \ldots, q(y_n, z_1, \ldots, z_n))$.

(r_1) $q(x, y, \ldots, y) \rightarrowtail y$;

(r_2) $q(x, e_1, \ldots, e_n) \rightarrowtail x$;

(r_3^i) $q(x, y_1, \ldots, y_{i-1}, x, y_{i+1}, \ldots, y_n) \rightarrowtail q(x, y_1, \ldots, y_{i-1}, e_i, y_{i+1}, \ldots, y_n)$;

(r_4^i) $q(x, y_1, \ldots, y_{i-1}, q(x, z_1, \ldots, z_n), y_{i+1}, \ldots, y_n) \rightarrowtail$
$q(x, y_1, \ldots, y_{i-1}, z_i, y_{i+1}, \ldots, y_n)$;

(r_5^i) If $x' < x$ then $q(x, y_1, \ldots, y_{i-1}, q(x', z_1, \ldots, z_n), y_{i+1}, \ldots, y_n) \rightarrowtail$
$q(x', q(x, y_1, \ldots, y_{i-1}, z_1, y_{i+1}, \ldots, y_n), \ldots, q(x, y_1, \ldots, y_{i-1}, z_n, y_{i+1}, \ldots, y_n))$

(r_6^i) If $x' < x$ then $q(x, y_1, \ldots, y_{i-1}, x', y_{i+1}, \ldots, y_n) \rightarrowtail$
$q(x', q(x, y_1, \ldots, y_{i-1}, e_1, y_{i+1}, \ldots, y_n), \ldots, q(x, y_1, \ldots, y_{i-1}, e_n, y_{i+1}, \ldots, y_n))$

Properties of the rewriting system

The rewriting system $\rightarrowtail$ is terminating and confluent.

For any formula Φ of any given tabular logic $\models_{(V,1)}$:

$$\models_{(V,1)} \Phi \quad \Leftrightarrow \quad \models_{(\mathbf{n},e_n)} \Phi^* \quad \Leftrightarrow \quad \Phi^* \rightarrowtail^* e_n$$

Rewriting binary decision diagrams

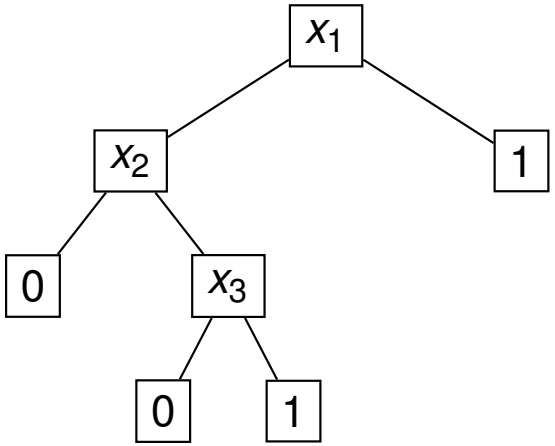
PC

$$\phi = x_1 \vee (x_2 \wedge x_3)$$

2-PC

$$\phi^* = q(x_1, q(x_2, 0, x_3), 1)$$

BDD



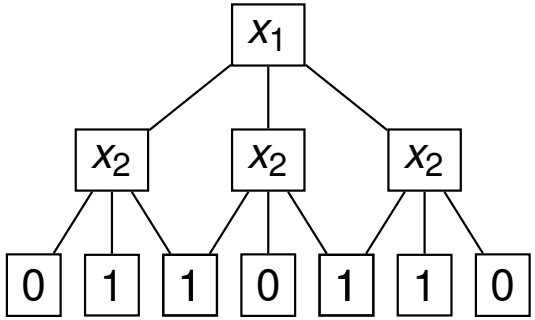
In the binary case, the rewriting system of head-normal 2-PC formulas is a reformulation of Zantema and van de Pol rewriting system for binary decision diagrams (2001).

Rewriting multi-valued decision diagrams

3-PC

$$\phi^* = q(x_1, q(x_2, 0, 1, 1), q(x_2, 1, 0, 1), q(x_2, 1, 1, 0))$$

MDD



Ongoing and further work

- The algebraic theory of n-BAs : Skew n-BAs, multi-ideals,. . .
- Proof theory of n-PC.
- Rewriting of multi-valued decision diagrams.

Thanks

Axioms and rules for the n-PC

$$\frac{}{\vdash_i e_i}\ (\texttt{Const}) \qquad \frac{\pi^{-1}(i) = \rho^{-1}(i)}{X^\pi \vdash_i X^\rho}\ (\texttt{Id})$$

$$\frac{\Gamma^{(i,j)} \vdash_i \Delta^{(i,j)}}{\Gamma \vdash_j \Delta}\ (\texttt{Sym}) \qquad \frac{\Gamma^{(i,j)} \vdash_i F, \Delta^{(i,j)} \quad i \neq k}{\Gamma, F^{(j,k)} \vdash_j \Delta}\ (\texttt{Neg1}) \qquad \frac{\sum_{i \neq j} \Gamma^{(i,j)}, F \vdash_i \Delta^{(i,j)}}{\Gamma \vdash_j F, \Delta}\ (\texttt{Neg2})$$

$$\frac{\sum_{j=1}^{n} \Gamma^{(j,i)}, F, G_j^{(j,i)} \vdash_j \Delta^{(j,i)}}{\Gamma, q(F, G_1, \ldots, G_n) \vdash_i \Delta}\ (\texttt{qL}) \qquad \frac{\sum_{j=1}^{n} \Gamma^{(j,i)}, F \vdash_j G_j^{(j,i)}, \Delta^{(j,i)}}{\Gamma \vdash_i q(F, G_1, \ldots, G_n), \Delta}\ (\texttt{qR})$$

$$\frac{\Gamma, F \vdash_i \Delta \quad \Gamma \vdash_i F, \Delta}{\Gamma \vdash_i \Delta}\ (\texttt{Cut}) \qquad \frac{\Gamma \subseteq \Gamma' \quad \Gamma \vdash_i \Delta \quad \Delta \subseteq \Delta'}{\Gamma' \vdash_i \Delta'}\ (\texttt{Weak})$$